

EQUIPMENT INSPECTION REPORT

Report No: IR-2024-0178

Date: 15 March 2024

Equipment: Heat Exchanger HX-301, Unit 3

Inspector: R. Kumar, Senior Inspector

FINDINGS:

1. Tube bundle shows external corrosion on 12 tubes (positions 3, 7, 15, 22, 28, 34, 41, 48, 52, 58, 63, 70). Wall thickness measured at 2.1mm (spec: 3.5mm minimum).
REPLACEMENT REQUIRED.
2. Shell-side nozzle N2 has weld seam crack extending 15cm circumferentially. Crack depth 4mm (spec: max 2mm).
REPAIR REQUIRED before next startup.
3. Gasket surface on channel head shows pitting corrosion, depth 0.8mm. MONITOR - acceptable for one more cycle.
4. Baffle plate 7 shows erosion thinning at edges.
Thickness 60% of original. REPLACE during next turnaround.

RECOMMENDATIONS:

- Priority 1: Replace tube bundle (estimated 3 weeks downtime)
- Priority 2: Repair N2 nozzle weld (estimated 1 week)
- Priority 3: Monitor gasket surface, replace at turnaround
- Priority 4: Replace baffle plate 7 at next turnaround

APPROVAL: Requires Plant Manager and Safety Officer sign-off before tube bundle replacement can proceed.